

Super-Li Inventory System

Operating Instructions & README

Version: adss2025_v02 | Module: Inventory

Software Systems Analysis & Design - Ben-Gurion University

1. System Requirements

Before running the system, make sure the following are installed:

Requirement	Version	Notes
Java (JDK)	11 or higher	Required to run the JAR (developed & tested on JDK 17)
Operating System	Windows / Mac / Linux	Any OS with Java support
Terminal / Command Prompt	Any	Used to run the JAR

No database server is needed - the system uses an embedded SQLite file that is created automatically on first run.

2. How to Run the System

Step 1 - Open a terminal and run the JAR file:

```
java -jar adss2025_v02.jar
```

Step 2 - The system menu is displayed. Choose the Inventory module:

```
=====
SUPERLI STORE MANAGEMENT SYSTEM
=====

1. Inventory Management Module
2. Human Resources (HR) Module
3. Exit System

Choose an option:
```

Step 3 - The Inventory module asks how to load its data:

```
=====

Welcome to Inventory Management
=====

1. Start empty
2. Load previous data from database

Choose:
```

Inventory Startup Option	Description
1. Start empty	Starts the Inventory module with no data; you add everything manually.
2. Load previous data from database	Loads the data already saved in the SQLite database (inventory.db). See section 12 for the data it currently contains.

There are no user accounts or passwords. All data is saved automatically to the SQLite file, so it persists between runs.

What's New in Assignment 2

- **Persistent SQLite database** - all data is saved locally to `inventory.db` and survives restarts. Choose startup option 3 to reload it.
- **Automatic supplier ordering** - when a removal drops a product below its minimum, the system automatically places a replenishment order through the supplier (mocked) and records a low-stock alert.
- **Warehouse entity** - warehouse stock is tracked as its own entity, kept consistent on every add / move / remove operation.
- **Incoming-stock tracking** - once an order is placed, the on-the-way quantity is counted so the system does not place a duplicate order before it arrives.

3. Main Menu Navigation

After startup, the main menu is displayed:

Option	Menu	Description
1	Category Operations	Add and view product categories
2	ItemType Operations	Manage product definitions (types)
3	Item Operations	Manage physical units of products
4	Discount Operations	Add and view promotions / discounts
5	Reports	Generate inventory and defective-item reports
6	Alerts	View low-stock alerts
0	Exit	Exit the system

4. Category Operations

Categories form a tree with unlimited levels. A category can have any number of sub-categories. Items may only be assigned to **leaf** categories (no children). Example: Dairy > Milk > 3%, where '3%' is the leaf.

Option	Action	Notes
1	Add root category	Creates a top-level category (e.g., Dairy)
2	Add sub-category	Creates a child category under an existing one
3	Show all categories	Lists all categories with full path
4	Show category by ID	Shows full details of a single category
5	Show root categories	Lists only top-level categories
6	Show category tree	Displays the full category hierarchy
0	Back	Return to main menu

5. ItemType Operations

An ItemType is a product definition (e.g., 'Milk 3% Tnuva 1L'). It holds the name, prices, store location and minimum stock threshold.

Option	Action	Notes
1	Add item type	Define a new product (name, prices, location, min)
2	Show all item types	Lists all product definitions
3	Show item type by ID	Shows full details of one product
4	Update minimum quantity	Change the low-stock threshold
5	Show item types by category	Filter products by category
0	Back	Return to main menu

Required fields when adding an item type:

Field	Description	Example
Name	Product name	Milk 3% Tnuva 1L
Shelf number	Physical shelf number in store	1
Aisle number	Physical aisle number in store	1
Min quantity	Alert threshold for low stock	20

Field	Description	Example
Cost price	Price paid to supplier	4
Selling price	Price charged to customer	6
Manufacturer	Name of manufacturer	Tnuva
Category	Leaf category from the tree	Dairy > Milk > 3%

6. Item Operations

An Item is a physical unit of a product. Each item tracks its location (shelf or warehouse), expiration date and damaged status.

Option	Action	Notes
1	Add single item	Add one physical unit with expiration date
2	Add multiple items of same type	Add N units at once with the same expiration
3	Show all items	Lists all physical units in the system
4	Show item by ID	Shows details of one unit
5	Show items by item type	Filter units by product type
6	Move one item to shelf	Move a warehouse item to the shelf
7	Move multiple items to shelf	Move N items of a type to the shelf
8	Move one item to warehouse	Move a shelf item to the warehouse
9	Mark item as damaged	Flag an item as damaged
10	Unmark item as damaged	Remove the damaged flag
11	Update item expiration date	Change the expiration date of an item
12	Remove item	Permanently delete a physical unit
13	Remove all defective items	Delete all damaged or expired units
0	Back	Return to main menu

Dates use the yyyy-mm-dd format (e.g., 2026-05-01). Expiration date is optional - leave blank and press Enter to skip.

Automatic Re-ordering (Assignment 2)

When removing items (option 12) or clearing defective items (option 13) drops a product's total quantity below its minimum threshold, the system automatically places a supplier order for the missing amount and shows a low-stock alert. The on-the-way quantity is tracked, so a second removal will not place a duplicate order while one is already pending.

7. Discount Operations

Two promotion types are supported: item-level and category-level discounts. If several discounts apply to the same item, only the highest percentage is used.

Option	Action	Notes
1	Add item discount	Apply a % discount to specific products
2	Add category discount	Apply a % discount to an entire category
3	Show active discounts for item	View discounts active today for a product
4	Show all discounts	List all discounts in the system
5	Show final price for item type	Calculate price after the best discount
0	Back	Return to main menu

When selecting several products or categories for a discount, type the listed Done / Cancel option to finish the selection.

8. Reports

Option	Report	Description
1	Category inventory report	All items per selected category with quantities and prices

Option	Report	Description
2	Show inventory by categories	Browse inventory filtered by one or more categories
3	Defective item report	Lists all damaged or expired items currently in the system
4	Purchasing report	Lists products below minimum with recommended order quantities
0	Back	Return to main menu

9. Alerts

Alerts are generated automatically when a product's total quantity (shelf + warehouse) drops below its minimum threshold.

Option	Action	Notes
1	Show all alerts	Lists all products currently below minimum quantity
2	Show alert for item type	Check if a specific product has a low-stock alert
0	Back	Return to main menu

10. Database Tables

All data is stored locally in one embedded **SQLite** database file shared by the integrated system, created automatically next to the JAR on first run. The complete list of tables in the database is below. The **Inventory** tables are owned and filled by this module; the **Employees / HR** tables belong to the partner module and share the same file.

Inventory module tables

Table	Rows	Stores
categories	36	Category tree (id, name, parent id)
item_types	21	Product definitions (name, location, prices, min, category, quantities)
items	87	Physical units (type, expiration, damaged flag, shelf / warehouse location)
discounts	3	Item and category discounts (type, percentage, start / end dates)
item_discount_targets	3	Links an item-discount to the item types it applies to
category_discount_targets	2	Links a category-discount to the categories it applies to
alerts	0	Low-stock alerts generated by the system

Employees / HR module tables (same shared database)

Table	Rows	Stores
employees	0	Employee records (name, bank details, employment type, salary, branch)
employee_roles	0	Roles each employee is qualified for
drivers	0	Driver licence type for employees who are drivers
availability	0	Weekly availability of each employee (morning / evening)
branches	0	Store branches (id, name, address)
shifts	0	Scheduled shifts (date, type, branch)
shift_assignments	0	Which employee is assigned to which shift and role
shift_required_roles	0	Roles (and amount) required for each shift

The database also contains the internal SQLite table `sqlite_sequence` (auto-increment bookkeeping). The unit / integration tests use a separate file (`inventory_test.db`) so they never affect the real data.

11. Tools & Libraries Used

Tools

Tool	Version	Purpose
Java (JDK)	17 (runs on 11+)	Programming language / runtime
Maven	3.x	Build & dependency management; produces the runnable JAR
SQLite	3 (embedded)	Local database engine (file-based, no server)

Modeling Tools

Tool	Purpose
draw.io (diagrams.net)	All UML diagrams - use case, class, sequence, and contracts

External Libraries

Library	Version	Purpose
sqlite-jdbc (org.xerial)	3.x	JDBC driver to read / write the SQLite database
JUnit Jupiter (junit-jupiter)	5.x	Unit & integration testing framework

Fill in the exact version numbers from your pom.xml (the <dependency> and <properties> sections) before submitting.

Running the tests

From the project root:

```
mvn test
```

20 tests run (15 from Assignment 1 + 5 new for Assignment 2: Warehouse entity, automatic ordering, no-duplicate-order, warehouse moves, and a SQLite persistence integration test).

12. Data Currently in the Database

The following is the data actually stored in the provided `inventory.db` (36 categories, 21 item types, 87 physical units, 3 discounts). It is loaded with the Inventory startup option 2. The stock levels are arranged to show every situation: products above the minimum, exactly at the minimum, below the minimum, empty, plus damaged and expired units.

12.1 Category Tree

Dairy (1)

Milk (2) > 3% (3), 1% (4), 5% (5)

Cheese (6) > Yellow (7), White (8)

Butter (9) > 250g (10), 500g (11)

Hygiene (12)

Shampoo (13) > 250ml (14), 500ml (15)

Soap (16) > Bar (17), Liquid (18)

Beverages (19)

Juice (20) > Orange (21), Apple (22)

Water (23) > 500ml (24), 1.5L (25)

Bakery (26)

Bread (27) > White (28), Whole Wheat (29)

Snacks (30)

Chips (31) > Regular (32), Baked (33)

Chocolate (34) > Dark (35), Milk Choc (36)

12.2 Item Types (Products)

ID	Name	Category	Cost	Price	Min	Manufacturer
1	Milk 3% Tnuva 1L	Dairy>Milk>3%	4	6	20	Tnuva
2	Milk 1% Tnuva 1L	Dairy>Milk>1%	4	6	20	Tnuva
3	Milk 5% Tnuva 1L	Dairy>Milk>5%	4	7	15	Tnuva
4	Yellow Cheese 200g	Dairy>Cheese>Yellow	8	12	10	Tnuva
5	White Cheese 200g	Dairy>Cheese>White	6	9	10	Tnuva
6	Butter 250g	Dairy>Butter>250g	6	9	8	Tnuva
7	Butter 500g	Dairy>Butter>500g	11	16	5	Tnuva
8	Pinuk Shampoo 250ml	Hygiene>Shampoo>250ml	8	14	8	Pinuk
9	Pinuk Shampoo 500ml	Hygiene>Shampoo>500ml	14	22	5	Pinuk
10	Dove Soap Bar	Hygiene>Soap>Bar	3	6	10	Dove
11	Dove Soap Liquid	Hygiene>Soap>Liquid	7	12	8	Dove
12	Prigat Orange Juice	Bev>Juice>Orange	4	7	15	Prigat
13	Prigat Apple Juice	Bev>Juice>Apple	4	7	15	Prigat
14	Water 500ml	Bev>Water>500ml	1	3	30	Neviot
15	Water 1.5L	Bev>Water>1.5L	2	5	20	Neviot

ID	Name	Category	Cost	Price	Min	Manufacturer
16	White Bread	Bakery>Bread>White	3	6	10	Berman
17	Whole Wheat Bread	Bakery>Bread>W.Wheat	4	7	10	Berman
18	Regular Chips 100g	Snacks>Chips>Regular	3	6	20	Elite
19	Baked Chips 100g	Snacks>Chips>Baked	4	7	15	Elite
20	Dark Chocolate 100g	Snacks>Choc>Dark	5	9	10	Elite
21	Milk Choc 100g	Snacks>Choc>Milk	4	8	12	Elite

12.3 Physical Items (Units) - actual stock in inventory.db

The starter stock is arranged so each scenario can be demonstrated. Valid units use the date 2027-03-01; expired units use 2026-04-25.

Item Type	Shelf	Warehouse	Total	Scenario
Water 500ml (14)	20	15	35	Well above minimum - removing one will NOT order
Milk 3% Tnuva 1L (1)	14	10	24	Above minimum
Yellow Cheese 200g (4)	6	4	10	Exactly at minimum - removing one triggers an order
Milk 1% Tnuva 1L (2)	3	2	5	Below minimum - restock = YES
Pinuk Shampoo 250ml (8)	2	1	3	Below minimum - restock = YES
Butter 250g (6)	4 + 1*	3	8	At minimum; *1 damaged - defective removal triggers an order
Whole Wheat Bread (17)	2	0	2	Expired units (2026-04-25)
All other products	0	0	0	Empty on purpose - far below minimum

* The Butter 250g shelf count includes 1 damaged unit (4 good + 1 damaged = 5 on shelf, 3 in warehouse, total 8).

12.4 Active Promotions

Type	Target	Discount	Start	End
Category	Dairy (1) - all sub-items	15%	2026-04-01	2026-04-30
Category	Hygiene (12) - all sub-items	10%	2026-04-15	2026-04-25
Item	Milk 3% / 1% / 5% (ids 1,2,3)	20%	2026-04-18	2026-04-20

End of Operating Instructions - Super-Li Inventory Management System - adss2025_v02